



TERRA LABORATORIES

Edge state estimation for closed living systems

How Terra Works

A technical brief for cultivation and operations teams.

The core idea: estimate the state you cannot measure

A grow room is a dynamical system. What actually decides the crop, the true conditions at the root, moves continuously, but your sensors sample it sparsely, indirectly, and with noise. Terra treats the root zone as a hidden state and estimates it directly: instead of reacting to a single probe crossing a threshold, it maintains a best estimate of the full internal state of the room and how confident it is in that estimate, updated every time new sensor data arrives.

The method

Online estimation	A recursive Bayesian estimator (an ensemble Kalman-type filter) fuses your live sensor streams with a mechanistic model of root-zone and room dynamics. Each step it produces a probability distribution over the hidden variables (root-zone O ₂ , true EC and pH, water-content behavior, disease pressure), not a single point value.
Offline calibration	The model is fit to your specific system and crop with hierarchical Bayesian calibration (Hamiltonian Monte Carlo) on your historical data. This is the bulk of what we do during a pilot, and it is why the engine reflects how your rooms actually behave rather than a generic template.
Probabilistic forecasting	The estimated state is propagated forward by Monte Carlo simulation to produce a calibrated probability of a developing failure over the next hours. Calibrated means that when it says seventy percent, it is right about seventy percent of the time. You get lead time with a confidence level, not a binary alarm.
Runs at the edge	The engine runs on a small node on site, next to the grow. It keeps working through network outages and does not require a room's data to leave your facility.

What it connects to

Terra reads the sensor streams you already run: substrate or solution EC and pH, water content or runoff, zone and canopy climate, and dissolved oxygen where you measure it. More signal helps, but the method is built to extract more from the sensors you already have rather than to require new ones. No rip and replace, and it does not take control of fertigation, dosing, or climate during a pilot.

What validation means here

We judge the engine two ways: calibration quality, meaning its stated probabilities match observed outcomes, and forecast lead time, meaning how many hours ahead it flags real events on data it was not trained on. In a pilot we report both against your own records, so the decision to scale rests on evidence from your rooms.

Honest status

These methods are established in state estimation and are validated in simulation for closed growing systems. What a pilot adds is calibration and proof on your real operation. We would rather show you lead time on your own data than ask you to take a benchmark on faith.

Next step: a short technical call to map this onto your rooms and sensor stack. Cameron Souza, Founder & CEO, Terra Laboratories. cameron@terralaboratories.com | terralaboratories.com